QUEZON CITY UNIVERSITY
COLLEGE OF COMPUTER STUDIES

WEEK 5-6
MANAGING SOFTWARE PROJECTS

SE101 – SOFTWARE ENGINEERING

TOPICS:

Managing Software Projects

- Project Management Concepts
- Process and Project Metrics
- Estimation for Software Projects
- Project Scheduling
- Risk Management (detailed discussion on Week 16)
- Maintenance and Reengineering

PROJECT MANAGEMENT CONCEPTS

Software Project Management refers to the branch of project management dedicated to planning, scheduling, resource allocation, execution, tracking and delivery of software and web project.

Software Project Roles and Responsibilities

Roles	Responsibilities
Project Manager	Develop a project plan, manage deliverables according to the plan, recruit project staff, Lead and manage the project team, Determine the methodology used on the project, Established a project schedule and determine each phase, assign task to project team members, provide regular updates to upper management
Functional Manager	Assign project, discuss how well person is doing the work and if person wants to continue doing it (providing opportunities for growth), gathering information from the other PMs to write the evaluation, work with member to set and coach on career goals
Operational Manager	Coordination and Supervision, Financial, HR, improve process and policies, coordinate and monitor the work of various department

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Software Project Roles and Responsibilities

Analyst	Ensuring that the requirements of the business client are captured and documented correctly before a solution is developed and implemented.
Change Manager	Accept changes to the project requirements, budget and timelines
Database Administrator	Models, designs and creates the databases and tables used by the software solution
Developer	Actual building of the solution
Tester	Ensures that the solution meets the business requirements and that is free of errors and defects
Client	Beneficiaries of a project or service.
User	Use the deliverables of the project.

PROCESS AND PROJECT METRICS

Software Management Activities

Project Planning

Scope Management

During Project Scope management, it is necessary to -

- Define the scope
- Decide its verification and control
- Divide the project into various smaller parts for ease of management.
- Verify the scope
- Control the scope by incorporating changes to the scope

Project Estimation

Project estimation may involve the following:

- Software size estimation
- Effort estimation
- Time estimation
- Cost estimation

For estimating project cost, it is required to consider -

- Size of software
- Software quality
- Hardware
- Additional software or tools, licenses etc.
- Skilled personnel with task-specific skills
- Travel involved
- Communication
- Training and support

Estimation for Software Projects

Project Estimation Techniques

- Decomposition Technique
  - Line of Code Estimation
  - Function Points Estimation
- Empirical Estimation Technique
  - Putnam Model
  - COCOMO

PROJECT SCHEDULING

Project Scheduling Activities

- Break down the project tasks into smaller, manageable form
- Find out various tasks and correlate them
- Estimate time frame required for each task
- Divide time into work-units
- Assign adequate number of work-units for each task
- Calculate total time required for the project from start to finish

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Resource Management

- Defining proper organization project by creating a project team and allocating responsibilities to each team member
- Determining resources required at a particular stage and their availability
- Manage Resources by generating resource request when they are required and de-allocating them when they are no more needed.

RISK MANAGEMENT

Project Risk Management

- Experienced staff leaving the project and new staff coming in.
- Change in organizational management.
- Requirement change or misinterpreting requirement.
- Under-estimation of required time and resources.
- Technological changes, environmental changes, business competition.

Risk Management Process

- Identification
- Categorize
- Manage
- Monitor

Project Execution and Monitoring

- Activity Monitoring
- Status Reports
- Milestones Checklist

Software Management Activities

Project Communication Management

- Planning
- Sharing
- Feedback
- Closure

Software Management Activities

Configuration Management

- Identification
- Validation
- Analysis
- Control Execution
- Close request

Project Management Tools

Project Management Tools – help and assist an individual or team to plan, organizing and managing work projects and tasks effectively from start to finish.

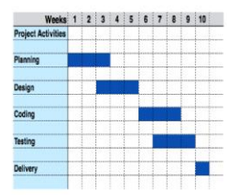
Tools used to manage the development process

- Gantt Chart
- PERT Chart / CPM
- Resource Histogram

Critical Path Analysis is useful in recognizing interdependent tasks in the project. It also helps to find out the shortest path or critical path to complete the project successfully.

Project Management Tools

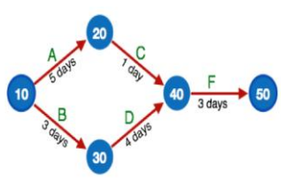
GANTT CHART was devised by Henry Gantt (1917). It represents project schedule with respect to time periods. It is a horizontal bar chart with bars representing activities and time scheduled for the project activities



Project Management Tools

PERT CHART (Program Evaluation & Review Technique) chart is a tool that depicts project as network diagram. It is capable of graphically representing main events of project in both parallel and consecutive way.

CPM (Critical Path Method) is a network analysis technique used in planning, scheduling and controlling of complex but routine activities

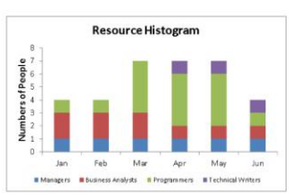


Project Management Tools

BASIS FOR COMPARISON	PERT	CPM
Meaning	PERT is a project management technique, used to manage uncertain activities of a project.	CPM is a statistical technique of project management that manages well-defined activities of a project.
What is it?	A technique of planning and control of time.	A method to control cost and time.
Orientation	Event-oriented	Activity-oriented
Evolution	Evolved as Research & Development project	Evolved as Construction project
Model	Probabilistic Model	Deterministic Model
Focuses on	Time	Time-cost trade-off
Estimates	Three time estimates	One time estimate
Appropriate for	High precision time estimate	Reasonable time estimate
Management of	Unpredictable Activities	Predictable activities
Nature of jobs	Non-repetitive nature	Repetitive nature
Critical and Non-critical activities	No differentiation	Differentiated
Suitable for	Research and Development Project	Non-research projects like civil construction, ship building etc.
Crashing concept	Not Applicable	Applicable

Project Management Tools

Resource Histogram - This is a an effective graphical tool for staff planning and coordination that contains bar or chart representing number of resources (usually skilled staff) required over time for a project event (or phase).



MAINTENANCE AND REENGINEERING

- **Software Maintenance**
  - It begins almost immediately after software is released to end users.
  - Maintenance corrects defects, adapts the software to meet a changing environment, and enhances functionality to meet the evolving needs of customers.

- **Reengineering**

  At a strategic level, **Business Process Reengineering** (BPR) defines business goals, identifies and evaluates existing business processes, and creates revised business processes that better meet current goals.

  Software reengineering encompasses inventory analysis, document restructuring, reverse engineering, program and data restructuring, and forward engineering. The intent of these activities is to create versions of existing programs that exhibit higher quality and better maintainability.

END OF PRESENTATION.
THANK YOU!